Predicting Employee Attrition Using Remote Work Productivity and Workplace Factors

GROUP PROJECT

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Course: CPSC 5071 – Data Management for Data Science

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Project Objective

Build a predictive model to identify factors associated with employee attrition by integrating workplace and remote productivity datasets.

The goal is to help organizations better understand attrition patterns and support data-driven retention strategies.

Problem Statement

Why Attrition Matters:

- Employee attrition
- High turnover
- Recruitment costs
- Training costs
- Productivity loss
- Job satisfaction
- Overtime workload
- Salary level
- Workplace conditions
- Remote work impact
- Employee productivity

Data and Data Management

Two datasets from Kaggle were used in this project

Dataset 1: Employee Attrition and Factors
(HR information such as salary, job satisfaction, overtime, and years at company)

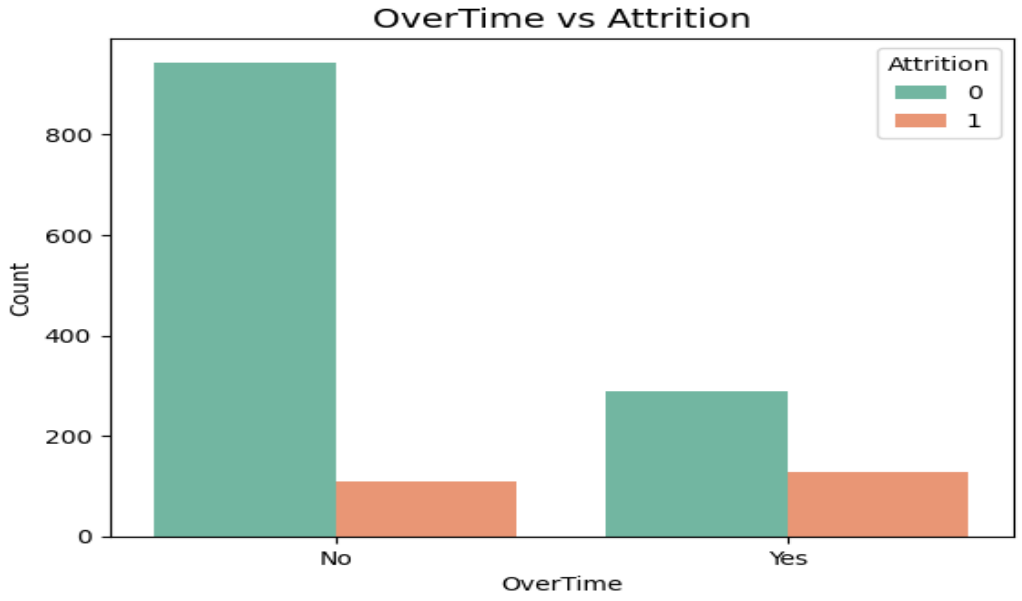
Dataset 2: Remote Worker Productivity Dataset
(productivity, stress level, and work environment factors)

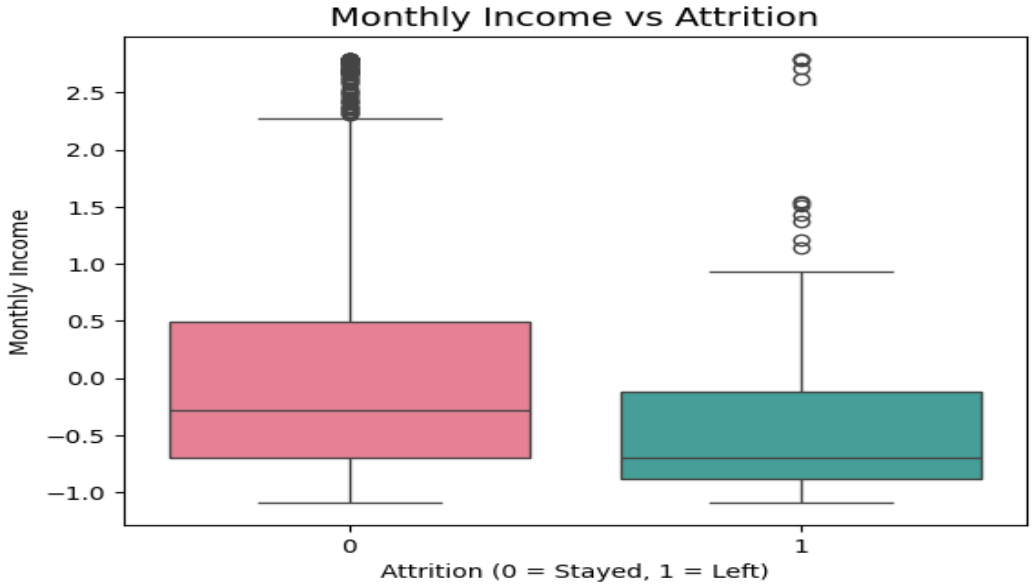
The datasets were integrated based on **experience level** to combine workplace and productivity information

Data preprocessing included cleaning, handling missing values, encoding categorical variables, and preparing the dataset for machine learning models

Exploratory Data Analysis

- Attrition distribution shows that most employees stay, creating class imbalance.
- Employees who work overtime tend to have a higher likelihood of leaving.
- Lower job satisfaction is associated with higher attrition rates.
- Income and years at the company also show noticeable differences between employees who stay and those who leave.





Modeling Approach

The dataset was divided using an 80/20 train–test split.

Cross-validation was applied during model training to improve generalization.

Two classification models were implemented:

- Logistic Regression
- Random Forest Classifier

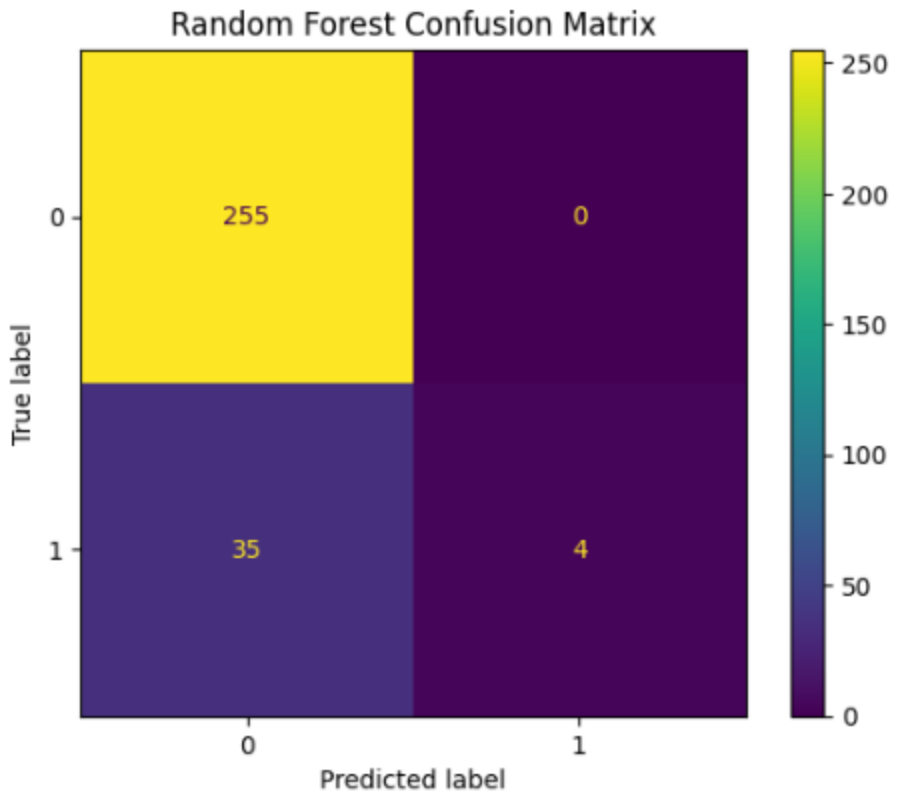
Model performance was evaluated using the following metrics:

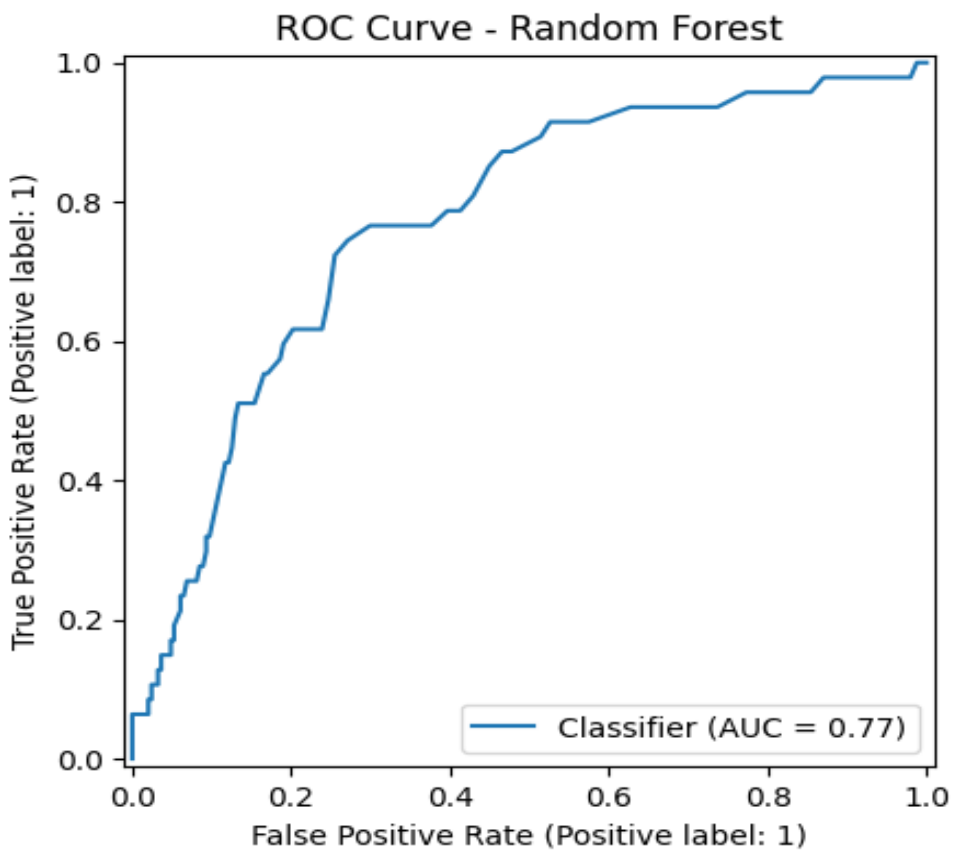
- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

The most influential factors identified by the models include:

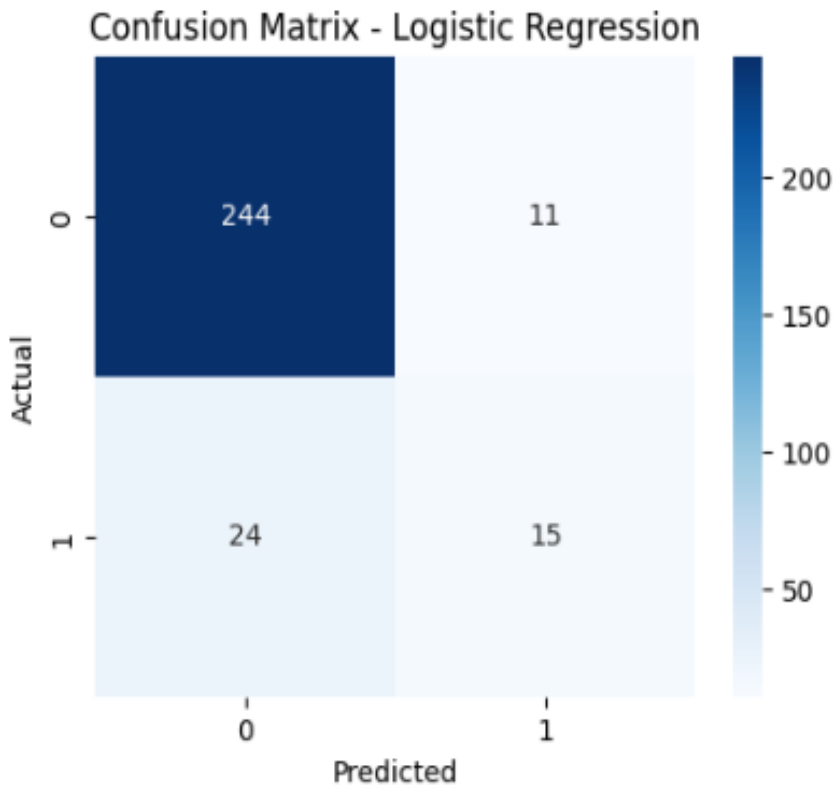
- Monthly Income
- Age
- Daily Rate
- Overtime
- Total Working Years
- Job Satisfaction
- Years at Company

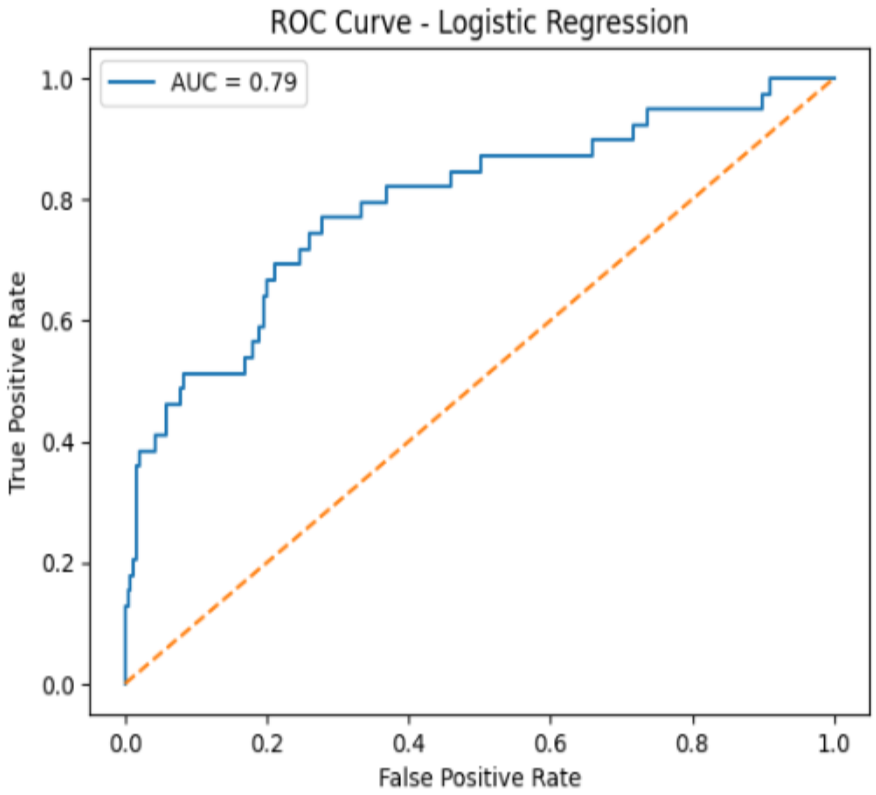
Random Forest





Logistic Regression





Model Results Comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.88	0.57	0.385	0.46	0.79
Random Forest	0.84	0.45	0.11	0.17	0.77

- Logistic Regression performed better across most evaluation metrics.
- It achieved higher accuracy, precision, recall, F1 score, and ROC-AUC.
- Random Forest struggled to detect employees who leave due to class imbalance in the dataset.

Therefore, Logistic Regression was the more reliable model for predicting employee attrition in this project.

Cross Validation and PR Curve

5-Fold Cross-Validation Results

Logistic Regression

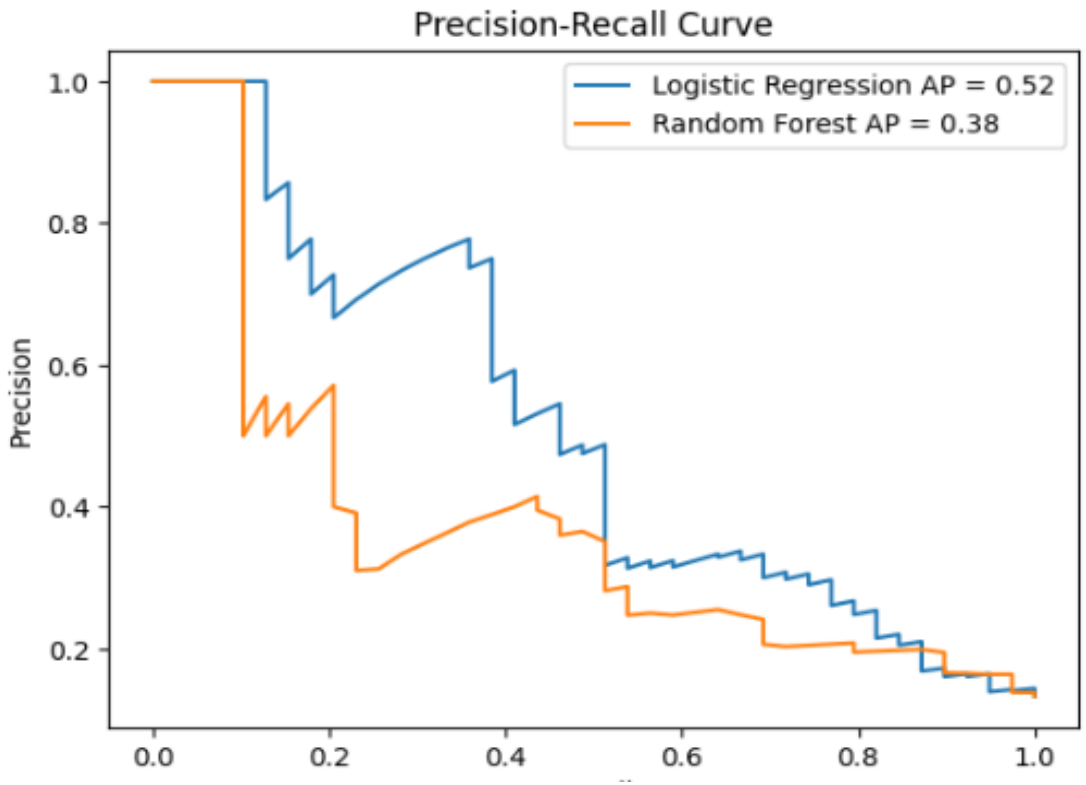
- AUC Scores (5 folds):
  - 0.858, 0.820, 0.867, 0.808, 0.865
- Mean AUC: 0.844
- Shows strong and consistent predictive performance

Random Forest

- AUC Scores (5 folds): 0.848, 0.781, 0.814, 0.773, 0.843
- Mean AUC: 0.812
- Performance is slightly lower than Logistic Regression

Key Insight

- Logistic Regression achieved higher average AUC
- Indicates better ability to distinguish employees who stay vs leave



Challenges and Future Work

Challenges in Predicting Employee Attrition

- **Class Imbalance:** Many more employees stayed than left → harder to predict attrition.
- **Limited Features:** Dataset lacks factors like engagement, work-life balance, leadership, and company culture.
- **Model Limitations:** Human behavior is complex; even accurate models may struggle to capture all drivers of attrition.

Future Work: Improving Employee Attrition Prediction

- **Advanced Models:** Explore Gradient Boosting, XGBoost, or Neural Networks for better performance
- **Handle Class Imbalance:** Use techniques like SMOTE or class weighting to better predict attrition
- **Feature Engineering:** Incorporate engagement surveys, performance metrics, and work-life balance indicators.
- **Real-Time Monitoring:** Develop a system to monitor employee risk and enable preventive actions

Conclusion

Employee Attrition Prediction: Model Comparison

- **Models Developed:** Logistic Regression & Random Forest
- **Training & Evaluation:** Same dataset used for fair comparison
- **Performance:**
  - Logistic Regression: Accuracy ≈88%, AUC 0.79
  - Random Forest: AUC ≈0.77
  - Logistic Regression performed better overall
  - Offers interpretability via coefficients → identifies key drivers (e.g., income, job satisfaction, tenure)

Logistic Regression is more suitable, balancing predictive performance and interpretability

Thank you